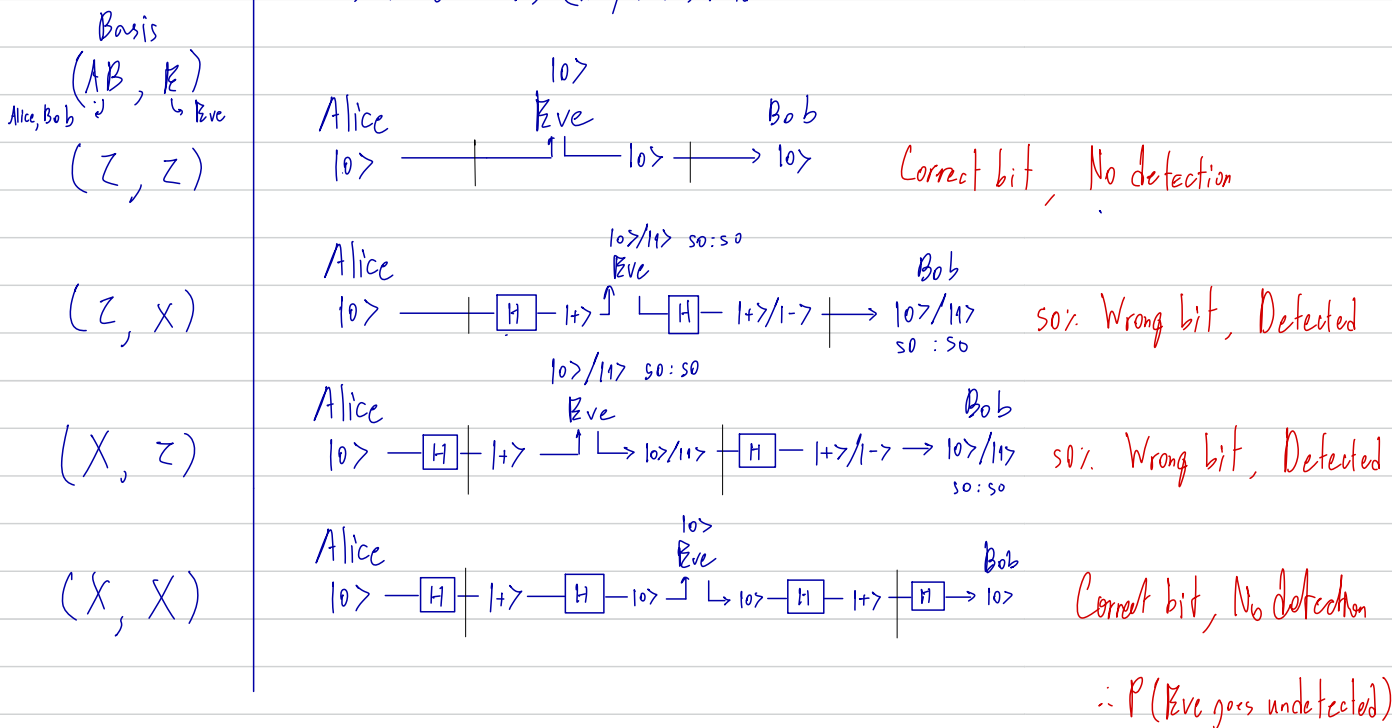


Z, X

Only consider when Alice, Bob chose the same basis (key bit is valid) * Eve will measure and retransmit in her chosen basis +



$$\therefore P(\text{Eve goes undetected}) = 0.75$$

Z basis: $|0\rangle, |1\rangle$
 $\uparrow \quad \uparrow$
 X basis: $|+\rangle, |-\rangle$

$$H|0\rangle = |+\rangle \Leftrightarrow H|1\rangle = |-\rangle$$

$$H|+\rangle = |0\rangle \Leftrightarrow H|-\rangle = |1\rangle$$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \rightarrow \text{When measure } |+\rangle \text{ (an X basis state) in Z basis}$$

$$\text{the result will be } \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} = 50\% |0\rangle, 50\% |1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \rightarrow \text{When measure } |0\rangle \text{ (a Z basis state) in X basis}$$

$$\text{the result will be } \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} = 50\% |+\rangle, 50\% |-\rangle$$

$$|1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$$

$|+\rangle$ is 0
 $|-\rangle$ is 1
 in the eye
 of X basis